Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/08/2013
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	The ISS Dynamic Lighting Schedule: An In-Flight Lighting Countermeasure to Facilitate Circadian Adaptation, Improve Sleep and Enhance Alertness and Performance on the International Space Station		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	slockley@hms.harvard.edu	**Fax:**	FY 617 732-4015
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-4977
Organization Name:	Brigham and Women's Hospital		
PI Address 1:	Division of Sleep Medicine and Circadian Disorders		
PI Address 2:	221 Longwood Avenue, Suite 438		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	10/01/2012	**End Date:**	09/30/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Czeisler, Charles Ph.D., M.D. (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HFP02801		
Performance Goal No.:			
Performance Goal Text:			

Space flight often exposes crews to unusual sleep-wake and work schedules that lead to misalignment of the circadian pacemaker, including abrupt 'slam-shifts' of work hours into the night, or entrainment to unusual 'day-lengths' (e.g., 24.65 h Martian sol), resulting in poor sleep, impaired alertness, and increased risk of fatigue-related accidents. Untreated circadian misalignment results in sleep and wake occurring at the incorrect circadian phase which reduces sleep quality and quantity and impairs alertness, reaction time, and cognition. Even without circadian misalignment, sleep duration is usually poor (~6h/night) during spaceflight and hypnotic medications and caffeine are commonly used to address insomnia and daytime sleepiness. In such a high-risk environment as the ISS, the risk of sleepiness-related performance decrements and accidents must be minimized. Light is a potential powerful countermeasure for both circadian misalignment and sleepiness associated with spaceflight. The effects of light are non-pharmacological and safe

Task Description:

and can be obtained from ambient lighting.
Recently, we and others have shown that short-wavelength (blue) light is most effective for phase-shifting the circadian pacemaker and enhancing alertness and performance. In addition to permitting vision, manipulation of short-wavelength blue light in particular has the potential to be a safe, non-pharmacological countermeasure to reduce the risk of circadian misalignment and performance deficits during space flight. These benefits can be achieved either by enhancing blue light to increase circadian resetting and alerting responses when required, for example when adapting to a 'slam-shift,' or depleting blue light to minimize the effects of light when these responses are not required, for example prior to sleep. Using blue light per se, however, is undesirable as it impairs visual function, an important consideration in spaceflight. Manipulation of the blue light content of white light has similar benefits, however, and provides a way to optimize both the visual and non-visual responses simultaneously.

A unique opportunity has arisen with the need to replace the current lighting aboard the International Space Station. NASA has proposed a new solid state lighting system with three pre-determined settings to address different operational needs: 1) white light for general vision; 2) blue-enriched white light to enhance high circadian adaptation and alertness; 3) blue-depleted white light to minimize alertness prior to sleep. We have developed a Dynamic Lighting Schedule which determines when each of these three settings will be used to optimize lighting to maintain visual function and as a countermeasure to facilitate circadian adaptation, improve alertness and performance, and enhance sleep. The current proposal will study how the new lights would be deployed to address the problems associated with a simulated slam shift, in a high-fidelity analog of the ISS lighting environment, sleep patterns, and work schedule. In a series of 8-day randomized clinical trials, we will test the hypotheses that deployment of the Dynamic Lighting Schedule in parallel with a gradual 8-hour phase advance shift in sleep-work schedules with improve the rate of circadian adaptation, improve sleep and improve cognitive performance, subjective alertness, and objective EEG correlates of alertness as compared to when the Dynamic Lighting Schedule is deployed during an abrupt 'slam' 8-hour shift of the sleep-wake and work schedule. Experiment 1 will assess the effect of continuous exposure to the DLS, whereas Experiment 2 will assess the effects of intermittent exposure. Experiment 1 is ongoing and to date 12 participants have completed the study, 6 per group (gradual versus 'slam'-shift). Data analyses are ongoing. Experiment 2 will begin once preliminary data are available from Experiment 1.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Light exposure is a requirement for all spaceflight and base missions and careful consideration is required to provide light of sufficient quality and quantity to ensure adequate vision and to optimize the circadian, neuroendocrine, and neurobehavioral effects of light, and ultimately crew health and safety. The requirement to replace the out-of-date fluorescent lighting onboard ISS has provided an opportunity to re-fit the ISS with the latest in solid-state lighting technology which has the capability to provide light of varying spectrum, intensity, and pattern. While ours and others' work has established blue and blue-enriched light as optimal to enhance circadian entrainment and alertness, final ground-based testing in high-fidelity analogs is required to develop guidelines for operational use prior to flight testing. The proposed study will provide the ground-based data necessary to inform guidelines for flight testing the new lighting source, in time for the fluorescent light replacement in 2015. The proposal specifically targets lighting countermeasures to address circadian misalignment and performance decrements during a 'slam-shift', which is a common requirement during Soyuz docking, and to enhance sleep and alertness during both slam-shift and normal operations. Without such data, future recommendations for operational deployment of the Solid State Lighting Assemblies (SSLAs) will be suboptimal. The Earth-based applications and commercialization potential of these studies are enormous. The 'slam-shifts' aboard ISS, and the consequences to sleep, circadian rhythms, performance, and health, are very similar to that experienced by the 15M Americans who do shiftwork. The World Health Organization recently designated shiftwork that involves circadian 'desynchrony' as a probable carcinogen and shift-workers also have high rates of night-time accidents and injuries, cardiovascular disease, and diabetes. Similar lighting interventions could be deployed for night shift-workers using the same Dynamic Lighting Schedule (DLS) which would have the benefit of improving productivity and safety while also reducing energy use by using targeted light spectra that only provide the light needed. Many non shift-workers also suffer circadian misalignment and performance problems; for example, the 'early-riser' workforce, school and college students, professions requiring long work hours, etc., would benefit from a similar DLS. Patients in hospitals and care homes are also often exposed to poor light-dark patterns, correction of which has recently been shown to slow the rate of cognitive decline in dementia. Anywhere where high levels of alertness, performance, and vigilance are required are target sites for such lighting interventions and provide enormous potential for commercialization.

Task Progress:

We began our recruitment efforts in Jan 2013 and to date we have screened 39 potential participants; 15 participants were excluded based on study inclusion/exclusion criteria, 12 participants have completed the study, and an additional four subjects are completing the screening portion of the study, waiting to start the inpatient phase of the study. One participant was disempanelled from the study. We expect to complete studies in 24 participants by the end of April 2014. All subjects complete an extensive screening procedure lasting about three weeks, which includes medical and psychological evaluations to determine eligibility. All participants studied have unremarkable physical, psychological, and ophthalmologic exams, including a negative Ishihara color blindness test. For at least three weeks prior to entering the Intensive Physiology Monitoring Unit at the Brigham and Women's Hospital, participants maintain a self-selected, constant 8 h sleep/rest/dark schedule confirmed with calls to a time- and date-stamped voicemail at bedtime and wake time for 3 weeks and with actigraphy for at least 7 days prior to entering the unit. Participants are asked to refrain from use of any prescription or nonprescription medications, supplements, recreational drugs, caffeine, alcohol, or nicotine. Compliance is verified by urine and blood toxicology during screening and urine toxicology upon entry to the unit. Participants are randomized to either the gradual or slam shift arms of the study on the day of admission. Participants then complete 8 days under the inpatient phase of the study which includes four cycles of dynamic light exposure protocols and a 30 hour constant routine protocol. The constant routine minimizes environmental oscillations that may act as time cues. For example, participants remain awake the whole duration in a semi-recumbent posture therefore negating the effects of sleep-wake cycles and postural changes, participants are fed frequent iscaloric snacks at regular intervals instead of daily meals negating the time cues of scheduled meals. This allows studying biologic rhythms under the control of the endogenous clock. The dynamic lighting is generated using the Solid State Lighting Module for Research (SSLM-R), a functional ISS lighting analog, which mimics both the geometry of the light sources aboard ISS and the light source and spectra that will eventually be deployed.

Bibliography Type:	Description: (Last Updated: 11/14/2025)
Abstracts for Journals and Proceedings	Rahman SA, St Hilaire MA, Flynn-Evans EE, Barger LK, Brainard GC, Czeisler CA, Lockley SW. "The ISS dynamic lighting schedule: An in-flight lighting countermeasure to facilitate circadian adaptation, improve sleep and enhance alertness and performance on the International Space Station." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Articles in Other Journals or Periodicals	Rahman SA, Flynn-Evans EE, Aeschbach D, Brainard GC, Czeisler CA, Lockley SW. "Diurnal spectral sensitivity of the acute alerting effects of light." Sleep. Accepted: 7/27/2013. , Jul-2013
Articles in Peer-reviewed Journals	Ruger M, St Hilaire MA, Brainard GC, Khalsa SB, Kronauer RE, Czeisler CA, Lockley SW. "Human phase response curve to a single 6.5 h pulse of short-wavelength light." Journal of Physiology (London). 2013 Jan 1;591(Pt 1):353-63. Epub 2012 Oct 22. http://dx.doi.org/10.1113/jphysiol.2012.239046 ; PubMed PMID: 23090946 , Jan-2013
Articles in Peer-reviewed Journals	St Hilaire MA, Sullivan JP, Anderson C, Cohen DA, Barger LK, Lockley SW, Klerman EB. "Classifying performance impairment in response to sleep loss using pattern recognition algorithms on single session testing." Accident Analysis and Prevention. 2013 Jan;50:992-1002. Epub 2012 Sep 5. http://dx.doi.org/10.1016/j.aap.2012.08.003 ; PubMed PMID: 22959616 , Jan-2013
Articles in Peer-reviewed Journals	Barger LK, Wright KP Jr, Burke TM, Chinoy ED, Ronda JM, Lockley SW, Czeisler CA. "Sleep and cognitive function of crewmembers and mission controllers working 24-h shifts during a simulated 105-day spaceflight mission." Acta Astronautica. 2014 Jan;93:230-42. Online first July 2013. http://dx.doi.org/10.1016/j.actaastro.2013.07.002 , Jan-2014
Articles in Peer-reviewed Journals	Barger LK, Sullivan JP, Vincent AS, Fiedler ER, McKenna LM, Flynn-Evans EE, Gilliland K, Sipes WE, Smith PH, Brainard GC, Lockley SW. "Learning to live on a Mars day: fatigue countermeasures during the Phoenix Mars Lander mission." Sleep. 2012 Oct 1;35(10):1423-35. PubMed PMID: 23024441 , Oct-2012
Articles in Peer-reviewed Journals	Gooley JJ, Ho Mien I, St Hilaire MA, Yeo SC, Chua EC, van Reen E, Hanley CJ, Hull JT, Czeisler CA, Lockley SW. "Melanopsin and rod-cone photoreceptors play different roles in mediating pupillary light responses during exposure to continuous light in humans." Journal of Neuroscience. 2012 Oct 10;32(41):14242-53. http://dx.doi.org/10.1523/JNEUROSCI.1321-12.2012 ; PubMed PMID: 23055493 , Oct-2012
Papers from Meeting Proceedings	Amundadottir ML, Lockley SW, Andersen M. "Simulation-based evaluation of non-visual responses to daylighting: Proof-of-concept study of healthcare re-design." 13th International Conference of the International Building Performance Simulation Association, Chambery, France, 2013 August 25-28. Proceedings of the 13th International Conference of the International Building Performance Simulation Association (IBPSA), Chambery, France, 2013 August 25-28. , Aug-2013
Papers from Meeting Proceedings	Amundadottir ML, St Hilaire MA, Lockley SW, Andersen M. "Modeling non-visual responses to light: Unifying spectral sensitivity and temporal characteristics in a single model structure." CIE Centenary Conference. Towards a new century of light, Paris, France, April 15-16, 2013. Proceedings of the CIE Centenary Conference. Towards a new century of light (CIE x038:20136), Paris, France, April 15-16, 2013. p. 101-110. , Apr-2013